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PANEL

Human Subjects Research in the Age of Generative AI: Opportunities and Challenges of Applying LLM-Simulated Data to HCI Studies

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human-AI collaboration, human-AI interaction, human-agent teamwork, AI for digital mental health, AI for future of work

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Abstract

Rapid advances in generative artificial intelligence suggest new possibilities for how human subjects research can be conducted in HCI studies. The panel invites both computer and social scientists to discuss future directions for applying simulated responses from large language models (LLM) for human subjects research. We discuss current challenges and opportunities in LLM simulations and brainstorm how insights across different disciplines might inform breakthroughs. We pay close attention to when and how applications of LLM simulations might augment human subjects research instead of steering it toward unintended directions. Discussions from the panel will provide preliminary ideas for when and how HCI researchers can apply LLM simulations to human subjects research pipelines. Through this engagement, we also aim to build a research community with shared interests.

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• **Human-centered computing → HCI design and evaluation methods**.

Keywords

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1 Background & Motivation

Human subjects research stands at the core of human-computer interaction (HCI) design and research. Recent advancements in generative artificial intelligence (AI) are poised to transform how these studies are conducted [7, 9], such as by adopting output generated from large language models (LLM) to simulate human responses, annotate data, and perform data analyses [8, 19]. In fact, many of these use cases have become prevalent [26]. Hence, ensuring the use of AI supports human subjects research rather than unintentionally skewing outcomes becomes a critical and timely challenge. In parallel, HCI and design communities have explored generative AI's potential for rapid prototyping and exploratory research (e.g., [21, 28]). Could similar approaches inform new paradigms of human subjects research? Addressing this question requires understanding both the technical capabilities of LLM performance and the complexities of conducting studies with human subjects.

Additionally, HCI scholars frequently draw theories and practices from outside fields to inform human research nuances. These include adopting insights from various social science fields, such as psychology, sociology, communication, and more, which possess a long history of investigating and comprehending human behaviors and experiences. How might established practices and literature in behavioral and social sciences inform effective applications of generative AI to human subjects research? The present panel seeks cross-disciplinary insights to address this inquiry.

This panel focuses on "LLM simulation of responses from human subjects research" (abbreviated as "LLM simulation" hereafter; [1, 11]) to ground the discussion on how generative AI might shape new frontiers for human subjects research. Through reviewing recent literature, we identify key technical challenges in LLM simulations, utilizing them as conversational starters to explore ***(1) how approaches to LLM simulations benefit from human research and social science insights and (2) how enhanced LLM simulation outcomes open up new ways to tackle long-standing, hard problems in human behavioral research.*** We select LLM simulation of human responses as a focal point for this panel for two key reasons: First, it is relevant to all HCI scholars and practitioners who conduct human subjects research. Besides, recent studies reveal the utility of accounting for practices in human research to improve LLM simulation outcomes (e.g., [10, 25]).

We expect the panel will elicit lively discussions and debates as scholars currently hold mixed opinions toward the use of LLM simulations for human subjects research. As the topic of LLM simulation garners increasing attention, prior events at CHI (e.g., [18]) and other academic conferences have initiated relevant discussions. This panel distinguishes itself in two key ways. First, ***it brings together computer scientists and social scientists to offer more holistic and diverse perspectives on the topic***. Previous events featuring scholars from the same discipline often elicited more uniform viewpoints. For example, a pre-conference event at SPSP'24 (The Society of Personality and Social Psychology, one of the largest academic conferences attended primarily by social and experimental psychologists) addressed a similar topic but was predominantly marked by opposition to the use of LLM simulation. In contrast, we aim to leverage the debate-driven nature of CHI panels, fostering dialogues that encompass both critical and supportive perspectives on the use of LLM simulations for human subjects research. Second, ***it adopts a problem-solving approach to guide the conversation***. Unlike prior discussions that primarily explored *whether* LLM simulations replicate human-like responses, this panel features researchers who have already demonstrated success with LLM simulations in their work. Building on top of their remaining challenges, the panel participants will discuss possible pathways toward *when* and *how* LLM simulations can be effectively applied to human subjects research.

2 Panel Format, Preparation, and Participation Plan

2.1 Before the Panel

We began preparing the panel by surveying the recent literature space, based on which we created a working document to actively exchange ideas and identify outstanding challenges of LLM simulations. In addition, the organizer met individually with panel participants (§4.1) and additional contributors (§4.2) to have in-depth conversations about their perspectives on the topic of LLM simulations. Together, we consolidate emerging themes and a list of questions for discussion (see more details in Section 3).

To ensure we cover the most important questions for HCI scholars, we will seek feedback from the broader CHI community to finalize the list of panel discussion questions. Specifically, we will distribute a short questionnaire through our professional networks and SIGCHI community channels, surveying HCI researchers' interest levels in our current discussion topics and questions while opening up the room for contributions beyond our own panel participants and contributors. We will kick off our panel with a brief report on the survey outcomes. Additionally, perspectives and questions that are deemed by our survey respondents will be emphasized during the panel discussion.

We will refer to the survey results to prioritize agenda items for the panel discussion and distribute a shortlist of discussion questions to our panel participants via an editable Google Doc. In that way, participants can start commenting on their opinions toward the shortlist of questions and attain preliminary ideas of others' perspectives prior to the panel. Additionally, we will host

organizing meetings to finalize the panel discussion details and logistics.

2.2 During the Panel

The panel will be conducted in a hybrid format as a roundtable discussion, led by two primary moderators and supported by the organizer. To ensure the panel discussion reflects both for-and-against standpoints regarding LLM simulation and its use in human subjects research, we assign informal roles among our panelists to either supportive or opposing voices regarding the use of LLM synthetic data. We will host the panel both on-site and via Zoom to accommodate participants and audiences who join either in person or remotely.

With the prepared list of discussion questions, the primary moderators will direct each question to panelists, guiding conversations with a semi-structured format:

> Present an outstanding challenge with applying LLM simulations for human subjects research
> → Prompt panelists who have done related work to elaborate on the nuance of the challenge
> → Invite other panelists and the audience to brainstorm how insights from existing theories and practices in human research could inform and improve LLM simulation approaches and how they are applied in human subjects research pipelines
> → Open up additional questions and comments from the audiences.

The organizer will monitor questions from both in-person and remote audiences, incorporating perspectives from the crowd to enrich the discussion while limiting off-topic engagement. The organizer will also take charge of time and content management, hinting the primary moderators when to move on from one question to another and whether important items are covered.

Throughout the panel, both on-site and remote audiences can use Mentimeter[1] to post their questions and comments. This allows audiences at different venues to interact with the panelists and other audiences. If we run short on time, the panelists and moderators can also respond to audiences' questions asynchronously via the Mentimeter platform.

A tentative schedule of the panel includes the following components: (1) Brief introductions of moderators and panelists (10 minutes); (2) Moderated roundtable discussion among panelists and the audience, led by the primary moderators (50 minutes); (3) Interactive Q&A session with the audience (15 minutes). If there is time left, we will ask each panelist to share their top-of-mind takeaway from the discussion as closing remarks.

2.3 After the Panel

The organizer will host a small, informal social event for panel participants (including moderators, panelists, and on-site audiences) to continue the conversation. This social event will most likely be in person at CHI and take the format of a lunch or dinner gathering. If there is substantial interest in continuing the discussion online, the organizer will host a Slack or Discord channel to provide a venue for knowledge and idea exchange after the panel.

Based on insights from the panel discussion, we plan to put together a position paper synthesizing the broader research community's interests, panelists' perspectives, and the audiences' responses to the topic. The organizer will lead and put together a draft of the paper and circulate it to all panelists and moderators to streamline and incorporate their feedback.

3 Panel Discussion Topics and Questions

As panels at CHI are not designed to delve into subjects comprehensively, we make our best attempt to narrow down the scope of discussion by focusing on three critical challenges in LLM simulations and their applications to human subjects research.

3.1 Challenge 1: Addressing homogeneity and biases of simulated output

Abundant work has revealed that LLM simulated output excels at capturing the means but yields a lower degree of variance of simulated human subjects' responses (e.g., [3, 17, 26, 27]). Besides, uniform outputs from LLM also tend to be biased in certain directions and can tend toward stereotypes of certain demographic groups. For instance, recent research experimenting with LLM to simulate public opinions found that simulated output tends to align more with liberal, wealthy, and educated demographics [5, 6, 20]. Given the homogeneity of LLM simulation outcomes, we will discuss key considerations HCI researchers should account for when incorporating LLM simulations into their studies. Possible discussion questions include:

- Which types and what topics of human subjects research might benefit more from LLM simulations?
- Alongside examining results from LLM simulations, when should researchers seek responses from a more diverse population and/or specialized subjects (e.g., expert participants)?
- Can existing social science and human behavioral theories inform when LLM simulated outcomes might be more diverse or uniform?
- To yield varying and realistic human responses, prior work has experimented with simulations that account for individuals' demographics and personas [2, 11]). What can we learn from established behavioral and social science literature in handling variables that capture individual differences?

3.2 Challenge 2: Scaling LLM simulations from the individual to group and network level

A substantial portion of human subjects research studies the social dynamics in small and large groups. However, the effect of group-level attributes on LLM simulations remains under-explored. Compared to simulating individual responses, simulations of group dynamics face more ecological challenges [13, 14]. Applying LLM to simulate network data, prior work found more accurate output with local instead of global approaches (i.e., simulating individual behaviors to form network data) [4]. The issue of homogeneity in

[1] Mentimeter (https://www.mentimeter.com/features/live-polling) is an online platform that allows for live polling and live chat. Both remote and on-site participants can scan a QR code to join a live chat. We chose this tool over live chat on Zoom so that in-person audiences can engage with those online without the need to log on to Zoom. This will greatly reduce technical glitches during the session (e.g., huge echoes when multiple participants unmute themselves on Zoom in the same physical space).

LLM-simulated outcomes are also seen in simulating structured networks, such that long-tailed degree distribution is absent in most simulated outcomes [4]. Besides, simulated networks demonstrated more extreme polarization, emphasizing political homophily, than real-world networks. Under this theme, possible questions for discussion include:

- Whether and how formal approaches to quantifying social influences can be incorporated in LLM simulation pipelines?
- Abundant social science literature discusses how social behaviors emerge in group settings. What can HCI scholars adopt from established work to improve simulation methods for groups and networks?

3.3 Challenge 3: Integrating LLM simulations to existing research pipelines

The last type of challenge concerns how HCI researchers practically adopt LLM simulations atop existing human subjects research protocols. Considering LLM simulations as a means to augment human subjects research, we discuss when human input (both researchers' expertise and participants' responses) is most valuable in the research process. Under this notion, possible discussion questions include:

- How do HCI researchers determine the extent to which they integrate LLM simulations in human subjects research, whether by simulating all responses, combining simulated and "real" responses, or relying entirely on human participants?
- What opportunities beyond formal data collection exist for using LLM simulations in human subjects research, such as for pilot studies, exploratory research, simulating missing data, or filling in underrepresented groups?
- How do HCI researchers effectively evaluate LLM-simulated outcomes beyond their resemblance to real human responses? How should these evaluative approaches adapt to different contexts, domains, and research purposes?

4 Panel Participants & Contributors

4.1 Panel Participants

Panel participants, including the organizer, moderators, and panelists, are those who will be present and join discussions with others and the audience at the panel. Here, we focus on elaborating on the related experience and role of each panel participant. Full bios of all authors are included in Appendix.

We would like to highlight that although the majority of panel participants are currently affiliated with U.S.-based institutions, everyone represents a unique combination of ethnicity, gender identity, and nationality. As mentioned above, we will distribute a questionnaire to gather scholars' opinions on the panel topic. We hope to leverage these pre-panel activities to ensure the panel discussion incorporates a wider range of perspectives – beyond those from our own panel participants.

Angel Hsing-Chi Hwang (Organizer attending in person; University of Southern California) researches human-AI collaboration through a wide range of human subjects research methods (Challenge 3). She initiated and organized the details of this panel. She also met individually with each panel participant and contributor to discuss their perspectives on LLM simulations. Given that she has the most holistic view of the full panel agenda, she will support the moderators and facilitate engagement from the audience at the panel.

Michael Bernstein (Moderator attending in person; Stanford University) leads a series of work on generative agents that simulate human behaviors and responses. Many of these projects (e.g., generative agents [14], social simulacra [15]) are by far some of the most recognized and oft-cited research on LLM simulations of humans. At the panel, Bernstein will lead discussions on the current challenges and opportunities of LLM simulations from technical perspectives.

S. Shyam Sundar (Moderator attending in person; Pennsylvania State University and Sungkyunkwan University) has extensive experience conducting behavioral experiments with human subjects. Sundar's experimental work led to some of the most referenced theoretical frameworks (e.g., the MAIN model [22], the TIME model [24], and its extension to human-AI interaction research [23]) for human subjects research in HCI. At the panel, Sundar will lead discussions on how existing behavioral and social science theories and practices can inform and improve LLM simulation approaches.

Manoel Horta Ribeiro (Panelist attending remotely; Princeton University) examines the impact of widespread LLM use on survey and online study responses conducted on platforms like Amazon Mechanical Turk [26]; many of these platforms are primary venues where HCI conduct human studies. At the panel, Ribeiro will discuss more in-depth the limitations of LLM simulations, specifically on the issue of high-quality but homogenous output (Challenge 1).

Serina Chang (Panelist attending remotely; Microsoft Research & University of California, Berkeley) explores the potential for LLM simulations of network data [4] and their applications for computational social science research (Challenge 2). At the panel, Chang will discuss open questions about simulating structured networks and bring in industry perspectives on the use of LLM simulations for research purposes.

Sherry Tongshuang Wu (Panelist attending in person; Carnegie Mellon University) studies the use of LLM simulations to replicate crowdsourcing results [30] and their broader implications for human-AI collaboration (Challenge 3). At the panel, Wu will discuss the utility of LLM simulations in crowdsourcing pipelines, which are applied by a substantial group of CHI researchers.

Renwen Zhang (Panelist attending in person; National University of Singapore) applies mixed methods and crowd research to study interpersonal interaction and human-machine communication. At the panel, Zhang will respond to the implications of LLM simulations of individual responses (as discussed by Ribeiro; Challenge 1) as well as the use of LLM simulations as tools in mixed methods research pipelines (Challenge 3).

Yingdan Lu (Panelist attending in person; Northwestern University) studies networks and cross-platform behaviors of human subjects through various computational social science approaches. At the panel, Lu will address future directions of human network research, given the current status quo of synthetic networks discussed by Chang (Challenge 2).

4.2 Additional Contributors

We seek input from additional contributors who help shape discussion topics and questions for the panel. Again, we underscore their related expertise and contributions in this section and present their full bios in Appendix.

Katya Ognyanova (Rutgers University) and **Aimei Yang** (University of Southern California) share their insights as social network scientists to help us establish the discussion theme on "Scaling LLM simulations for group and network data" (Challenge 2). Additionally, **Dmitri Williams**' (University of Southern California) research on social values (SV) [29] and quantification of social influences [31] informs specific discussion questions on whether and how effectively accounting for the impact of social values on individual behaviors in the prompting and in-context learning process might improve the performance of LLM simulated group and network dynamics (Challenge 2).

Joon Sung Park (Stanford University) works first-hand on various highly recognized studies of generative agents and LLM simulations work [14–16]. Such insights inform specific contexts where LLM simulations see either more potential or limitations (Challenge 1). Additionally, Park and **Aaron Shaw** (Northwestern University) are part of a larger research team that recently attempted simulations of human responses from 1,000 subjects [16]. We also seek insights from Shaw, who has done extensive research on social computing and crowd platforms to consolidate discussion questions related to Challenge 3.

Ziang Xiao's (Johns Hopkins University) ongoing work examines the use of AI tools in social science research, including designing agents that can elicit more genuine responses from participants. Building on his previous research on evaluative approaches for language models [12], Xiao posed critical questions regarding the evaluation of LLM simulations (Challenge 3).

David A. Shamma (Toyota Research Institute) advises the panel participation plan based on his insights as a former Technical Program Chair at CHI and his past experiences of running numerous successful panels at the conference.

5 Expected Outcomes & Contributions to the CHI Community

This panel seeks to inspire the CHI community to critically explore the effective and rigorous use of LLM simulations in human subjects research. Moving beyond debates about how closely LLM simulations mirror actual human responses, we recognize the growing presence of simulated data and responses and focus on how HCI scholars can respond to this rapid transformation By engaging in this discussion, we aim to develop preliminary frameworks that guide the integration of LLM simulations into existing research pipelines. Additionally, we aspire to foster the emergence of a research community united by a shared interest in advancing this topic.

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Appendix

A Full Bios of Panel Participants & Additional Contributors

Angel Hsing-Chi Hwang is an Assistant Professor at USC Annenberg and a human-AI interaction (HAII) researcher. Her research explores the societal impact of artificial intelligence (AI) on work practices in various applied domains, such as small-group teamwork, content production, and mental healthcare services. Hwang applies a wide range of methods to research these topics, ranging from behavioral experiments to computational content analysis and qualitative methods. Hwang received her PhD in communication at Cornell University with a concentration in human-computer interaction and conducted her postdoctoral training at Cornell Bowers College of Computing and Information Science.

Aimei Yang is an Associate Professor at USC Annenberg. Yang earned her PhD in communication at the University of Oklahoma; her research is positioned at the intersection of strategic public relations research and a social network approach to the study of interorganizational relationships, and focuses on the relationship dynamics among civil actors, corporations and government agencies. Specifically, Yang's research topics include transnational civil networks, multisectoral alliance networks, public diplomacy, and dynamic social networks in social movements.

S. Shyam Sundar is Evan Pugh University Professor and James P. Jimirro Professor of Media Effects. He is the founding director of the Media Effects Research Laboratory at Penn State University's College of Communications. He is also Director of the university-wide Center for Socially Responsible Artificial Intelligence. He currently holds a visiting appointment as Distinguished Professor in the Department of Immersive Media Engineering at Sungkyunkwan University in South Korea, supported by MSIT (Ministry of Science, ICT), Korea, under the Global Scholars Invitation Program (RS-2024-00459638) supervised by IITP (Institute for Information & Communications Technology Planning & Evaluation).

Renwen Zhang is an Assistant Professor in the Department of Communications and New Media at the National University of Singapore (NUS). Her research examines the role of digital technologies in mediating and shaping interpersonal communication and well-being, with a particular focus on social support, self-disclosure, and mental health. Her recent work focuses on AI-mediated supportive communication and human-machine communication. Dr. Zhang's work is largely interdisciplinary and has been published in top-tier journals and conference proceedings in Communication and HCI, including the Journal of Computer-Mediated Communication, CHI, and CSCW.

Joon Sung Park is a PhD student in computer science at Stanford University. Park's work envisions new forms of human-computer interaction by leveraging state-of-the-art advances in natural language processing and machine learning. In particular, this work introduced the concept of, and the techniques for creating generative agents–computational agents that simulate human behavior. Park has been recognized with best paper awards at UIST and CHI, as well as nominations at CHI, CSCW, and ASSETS, and has been reported in venues such as The Times, The Guardian, NBC, The New York Times, The New Yorker, Forbes, Wired, Science, and Nature.

Manoel Horta Ribeiro is an incoming Assistant Professor at Princeton University in the Department of Computer Science, affiliated with the Princeton Center for Information Technology Policy (CITP). Ribeiro works at the intersection of computational social science, artificial intelligence, and social computing, tackling real-world issues and providing actionable insights for policy and design changes. Ribeiro's research has also received extensive media coverage, helping shape the public debate around online platforms and generative AI.

Yingdan Lu is an Assistant Professor in the Department of Communication Studies at Northwestern University. She co-directs

the Computational Multimodal Communication Lab and is affiliated with the Center for Communication & Public Policy and the Center for Human-Computer Interaction + Design. Lu's research applies cutting-edge computational methods to large-scale multimodal datasets to understand the evolution of digital propaganda in authoritarian regimes and how individuals encounter and communicate multimodal (mis)information in AI-mediated environments. Her work has appeared in leading peer-reviewed journals across communication, political science, and human-computer interaction.

Serina Chang is an incoming Assistant Professor at UC Berkeley, with a joint appointment in EECS and Computational Precision Health. She recently completed her PhD in CS at Stanford University and is currently a postdoc at Microsoft Research NYC in the Computational Social Science group. Chang's research falls at the intersection of AI, public health, and social science. Specifically, she develops AI and graph methods to study human behavior, improve public health, and guide data-driven policymaking. Her work has been recognized by a KDD Best Paper Award, NSF Graduate Research Fellowship, Meta PhD Fellowship, Stanford Finch Family Fellowship, EECS Rising Stars, Rising Stars in Data Science, and Cornell Future Faculty Symposium, and has been featured by over 650 news outlets, including The New York Times and The Washington Post.

Sherry Tongshuang Wu is an Assistant Professor in the Human-Computer Interaction Institute at Carnegie Mellon University (also affiliated with the Language Technology Institute). Her research lies at the intersection of Human-Computer Interaction and Natural Language Processing, aiming to support humans interacting with imperfect AIs, by debugging and correcting AIs interactively. She has authored award-winning papers in top-tier NLP, HCI and Visualization conferences and journals such as ACL, CHI, TOCHI, TVCG, etc. Before joining CMU, Sherry received her Ph.D. degree from the University of Washington and her bachelor degree from the Hong Kong University of Science and Technology, and has interned at Microsoft Research, Google Research, and Apple.

Katherine Ognyanova is an Associate Professor in the School of Communication and Information at Rutgers University. Her work examines the effects of social influence on civic and political behavior, confidence in institutions, information evaluation, and public opinion formation. Her methodological expertise is in computational social science, network science, and survey research. Ognyanova is the director of the Rutgers Computational Social Science Lab. She is also one of the founders and a principal investigator for the COVID States Project and the Civic Health and Institutions Project: two large multi-university initiatives exploring public attitudes to politics and health.

Dmitri Williams is a Professor at USC Annenberg, where he teaches courses on technology and society, games and data analytics. His current work focuses on the study of influence among populations through the concept of "social value." His ongoing work centers on the social and economic impacts of new media, often within online games. He works actively with companies and startups across the tech sectors. Williams was the first researcher to use online games for experiments and to undertake longitudinal research on video games, and now uses a wide variety of methods ranging from experiments, surveys, and machine learning to content analysis and the occasional ethnography.

Michael Bernstein is an Associate Professor of Computer Science at Stanford University, where he is a Bass University Fellow and Interim Director of the Symbolic Systems program. His research focuses on designing social, societal, and interactive technologies. This research has been reported in venues such as The New York Times, Wired, and TED AI. Michael has been recognized with an Alfred P. Sloan Fellowship, the UIST Lasting Impact Award, and the Computer History Museum's Patrick J. McGovern Tech for Humanity Prize. He holds a bachelor's degree in Symbolic Systems from Stanford University, as well as a master's degree and a Ph.D. in Computer Science from MIT.

David A. Shamma is a distinguished industry scientist researching people, artificial intelligence, and media experiences. His current research focuses on Carbon Neutrality at Toyota Research Institute's Human-Centered AI division. He has worked on Edge AI and future of work at FX Palo Alto Laboratory, AI sensors for wearable fashion at Centrum Wiskunde & Informatica (CWI), HCI + AI as a director of research at Yahoo Labs/Flickr, and UXR for remote knowledge sharing at NASA's Center for Mars Exploration. He is a Distinguished Member of the ACM, a Senior Member of IEEE, and holds over 30 patents. Ayman holds a Ph.D. in Computer Science from Northwestern University and completed his M.S./B.S. at the IHMC at UWF.

Aaron Shaw is an Associate Professor of Communication Studies and Sociology (by courtesy) at Northwestern University, where he is affiliated with the Center for Human-Computer Interaction + Design, the Institute for Policy Research, and the Buffett Institute for Global Affairs. Shaw is also a Faculty Associate of the Berkman-Klein Center for Internet & Society at Harvard University; co-founder of the Community Data Science Collective. He has held a fellowship at the Center for Advanced Study in the Behavioral Sciences (CASBS) at Stanford University; and will serve as "Scholar in Residence" in King County, Washington during the 2022-2023 academic year.

Ziang Xiao is an Assistant Professor in Computer Science at Johns Hopkins University. Xiao's research is motivated by the fundamental question of understanding humans at scale. Their goal is to study human-computer interaction to expand our knowledge about our behaviors and decision-making processes. Currently, Xiao's research focuses on three exciting topics, AI for Social Science, Human-centered Model Evaluation, and Information Seeking. Broadly, their work lies in the intersection of human-computer interaction, natural language processing, and social and personality psychology.